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Beyond the Canvas: Using Machine Learning to Detect Psychological Signals in Children's Drawings

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1. Introduction

Mental health assessment in children is a critical yet challenging domain in clinical psychology. Traditional evaluation methods rely heavily on direct interviews, standardized questionnaires, and observations conducted by trained professionals, approaches that are time-consuming, resource-intensive, and often ineffective with young children who may lack the verbal skills to articulate their inner emotional states. Early detection of psychological distress in children is vital, untreated mental health issues at a young age can lead to developmental delays, academic difficulties, and long-term psychiatric conditions. Therefore, There is a pressing need for non-intrusive, scalable, and objective tools that can support clinical evaluation.

Psychologists have observed that features within drawings can carry diagnostic significance. Dark, chaotic imagery, distorted human figures, or extreme isolation of drawn subjects are among the indicators that trained clinicians associate with potential psychological distress. However, systematic and reproducible interpretation of these features at scale remains difficult with human analysis alone.

This project explores whether supervised machine learning — specifically image classification techniques — can automate and standardize the analysis of children's drawings as a psychological screening tool. The goal is to determine whether a model trained on annotated drawing samples can learn to extract and combine visual features in ways that correlate meaningfully with emotional states. Such a model could serve as a low-cost, scalable first-pass screening tool that flags children who may benefit from further evaluation, particularly in settings with limited access to trained psychologists.

The intended task is formulated as a supervised multi-class image classification problem. The input to the system is a raw digital image of a child's drawing (224×224 pixels, RGB format) submitted without any metadata. The output is a discrete categorical label representing one of four emotion classes: Happy, Angry, Sad, or Fear, as defined by the dataset annotation scheme. The model is expected to learn visual representations that capture psychologically relevant features without explicit hand-engineering of those features. Success

will be measured by classification accuracy, F1-score (given potential class imbalance), and agreement with human expert labels. Figure 1 below illustrates the overall pipeline.

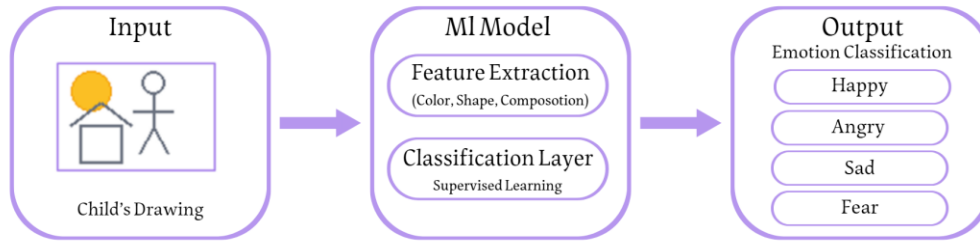


Figure 1: Input \ Output

2. Related Works

Several prior works have explored the intersection of deep learning, visual art analysis, and child psychological assessment. The following four studies are most closely related to the current project.

- **Emotion Analysis of Children's Drawings: [1]**

This study targeted binary emotion classification (Happy vs. Sad) in children's hand-drawn artwork. They introduced the KIDO dataset of 10,860 scanned drawings from 5,430 students. Image-based models (DenseNet, ResNet, ConvNeXt, ViT, Swin) achieved a macro F1 of up to 0.79; text-based classifiers (BERT) trained on children's written reflections exceeded 97% accuracy. This work confirmed that CNNs can learn discriminative features from children's drawings, directly motivating our CNN-based approach.

- **AI-Driven Classification of Children's Drawings for Pediatric Psychological Evaluation: An Ensemble Deep Learning Approach: [2]**

This study used the KHMD dataset — children aged 5–10 tracing shapes on a custom app, labeled 'normal' or 'not normal' — and applied transfer learning with VGG16, VGG19, and MobileNet combined via hard-voting ensemble, achieving 99% accuracy. This study motivated

our use of VGG16 and ensemble methods and confirmed that transfer learning is effective even on small, specialized drawing datasets.

- **Emotion Recognition in Visual Art using Deep Learning Frameworks (2021) [3]**

This study developed a multimodal CNN framework for the ArtEmis dataset of fine art paintings annotated with emotion labels and natural language descriptions. Their model exceeded 60% accuracy in multi-class emotion classification, validating deep learning for affective visual content beyond photography.

- **Visual Sentiment Analysis in Art using D-ViSA Dataset (2023) [4]**

This study introduced a framework for detecting emotional dimensions in abstract art images, focusing on continuous rather than discrete emotional representations. The D-ViSA dataset of abstract paintings annotated with categorical emotions and dimensional valence/arousal scores was used. Deep learning models were trained for both classification and regression tasks, achieving correlation scores of 0.6–0.7 in valence and arousal prediction, extending the basis for computational emotion recognition in non-representational visual content.

3. Data

3.1. Data Description

The Children’s Drawings Emotion Dataset is a collection of RGB images depicting drawings created by children and categorized according to their associated emotional states. The dataset consists of four emotion classes: **happy, angry, fear, and sad**. Each image represents a child’s drawing that visually expresses one of these emotional categories.

The dataset contains a total of 407 images, distributed as follows: 100 images labeled as happy, 100 as angry, 101 as fear, and 106 as sad. The relatively balanced distribution across

classes reduces the risk of bias during model training and allows for fair multi-class classification evaluation.

All images are colored drawings with varying resolutions and artistic styles, reflecting authentic children's visual expression patterns. Prior to model training, images will be resized to a fixed dimension (e.g., 224×224 pixels) to ensure compatibility with convolutional neural network architectures. Since the dataset size is limited, data augmentation techniques such as random rotation, horizontal flipping, scaling, and brightness adjustment will be applied to increase variability and reduce overfitting.

The dataset was obtained from Kaggle under the title Children's Drawings Emotion Dataset. It was selected because it directly supports the objective of developing a supervised learning model for emotion classification in children's drawings. The labeled emotional categories enable a clear multi-class classification task, while the visual diversity of the drawings provides meaningful patterns for model learning and analysis.

3.2 Data Summary Statistics

Attribute	Description
Total Images	407 (resized to 224×224 pixels for training)
Number of Features	RGB pixel values of the images
Number of Classes	4 (happy, angry, fear, sad)
Class Distribution	Happy: 100, Angry: 100, Fear: 101, Sad: 106
Dataset Split	Training, Validation, and Test sets
Data Type	Colored children's drawings (RGB images)

Table 1: Data Summary Statistic

3.3 Representative Examples from the Dataset

Class	Image	File Name
Angry		a2.jpg
Happy		h2.jpg
Sad		s9.jpg
Fear		f17.jpg

Table 2: Representative Examples from the Dataset

3.4 Data Preprocessing

All images were resized to 224×224 pixels and pixel values were normalized to [0,1] by dividing by 255. Prior to splitting, the original 407 images were offline-augmented by applying flips and rotations directly to the raw images, expanding the dataset to 943 images to improve model generalization. The expanded dataset was then split per class using a stratified 70/15/15 scheme (train / validation / test) with random seed 42 via scikit-learn's train_test_split.

Additional data augmentation was applied to the training set only using Keras ImageDataGenerator, with the following techniques: horizontal flipping, rotation up to $\pm 20^\circ$,

width/height shifts up to 20%, zoom up to 20%, shear of 10%, and brightness adjustment in [0.8, 1.2]. The validation and test generators used only rescaling — no augmentation — to ensure unbiased evaluation. For the SVM baseline, a separate pipeline was applied: images were converted to grayscale, resized to 64×64 pixels, and HOG features were extracted before normalization.

4. Methods

Five models were implemented and compared across two families: one classical machine learning baseline and three deep learning models combined into an ensemble. This design allows a direct comparison of handcrafted feature engineering versus learned deep representations for this task.

4.1 SVM with HOG Features (Classical Baseline)

An SVM with an RBF kernel is used as the baseline to quantify what deep learning adds over traditional methods. SVMs find an optimal decision boundary in a high-dimensional feature space using kernel functions to handle non-linear separability. Features are extracted using the Histogram of Oriented Gradients (HOG) descriptor, which captures the distribution of local edge directions making it sensitive to the stroke patterns and contours that define emotion in drawings (e.g., upward curves for happiness, downward strokes for sadness). HOG is computed on 64×64 grayscale images with 9 orientation bins, 8×8 pixels per cell, and 2×2 cells per block, yielding a 1,764-dimensional feature vector per image. Features are then standardized using StandardScaler before classification.

4.2 Transfer Learning CNNs

Three ImageNet-pretrained CNN architectures are fine-tuned on the drawings’ dataset. Transfer learning is justified by the small dataset size (407 images) pretrained models provide rich visual representations that cannot be learned from scratch at this scale. All three models share

an identical custom classification head appended to each pretrained base: GlobalAveragePooling2D → BatchNormalization → Dense (256, ReLU) → Dropout(0.4) → Dense(128, ReLU) → Dropout(0.2) → Dense(4, Softmax). Dropout layers provide regularization against overfitting.

- **MobileNetV2** (~3.4M parameters) uses depthwise separable convolutions and inverted residual blocks, making it the lightest and most computing efficient of the three. It is included to test whether a compact architecture can still learn meaningful emotion features from drawings.
- **ResNet50** (~25M parameters) is a 50-layer residual network that uses skip connections to prevent vanishing gradients in deep architectures. It is a proven general-purpose feature extractor on image classification benchmarks and provides a strong reference point for comparison.
- **VGG16** (~138M parameters) is a classic deep CNN with 13 convolutional layers all using 3×3 filters, followed by the custom head. Its high capacity and sequential structure make it well-suited for learning the texture and stroke-based features present in children's drawings.

4.3 Soft-Voting Ensemble

The ensemble averages the probability vectors from all three CNN models: $P_{\text{ensemble}} = (P_{\text{MobileNetV2}} + P_{\text{ResNet50}} + P_{\text{VGG16}}) / 3$, with the final class determined by argmax. Soft voting is preferred over hard voting because it incorporates each model's confidence, giving more weight to higher-confidence predictions in cases of disagreement.

5. Experiment

5.1 Setup, Data Preparation, and Augmentation

All experiments were run on Google Colab with a GPU (TensorFlow 2.19.0, confirmed output: GPU: [PhysicalDevice]). A global random seed of 42 was set across Python, NumPy, and TensorFlow for reproducibility.

The following libraries were used: TensorFlow/Keras (model building and training), scikit-learn (data splitting, SVM, GridSearchCV, and evaluation metrics), scikit-image (HOG feature extraction), Pillow (image loading and resizing), pandas (results tabulation), matplotlib and seaborn (visualizations). All experiments were run on Google Colab with a GPU accelerator (TensorFlow confirmed GPU availability at runtime), which significantly reduced CNN training time compared to CPU-only execution.

The dataset loaded from Google Drive contained 943 images across four classes (the original 407-image Kaggle dataset had been pre-augmented in the Drive folder before splitting). These were split per class using scikit-learn's `train_test_split` with stratification and `random_state=42`, using a 70% train / 15% validation / 15% test ratio. Stratification ensures each split maintains the same class proportions as the full dataset.

All images were resized to 224×224 pixels, and pixel values were normalized to $[0, 1]$ by dividing by 255 consistent with the preprocessing used during original ImageNet training of the pretrained CNNs. Keras ImageDataGenerator applied on-the-fly augmentation per batch to the training set only; the validation and test generators used plain rescaling, so evaluation metrics reflect performance on unmodified images. The table below lists the augmentation techniques and the reasoning behind each value.

Technique	Value	Why This Value
Horizontal Flip	True	Mirroring does not change emotion valid transformation for all classes

Rotation Range	$\pm 20^\circ$	Simulates different paper orientations without distorting emotional meaning
Width / Height Shift	$\pm 20\%$	Simulates drawings not centered on the page, common in children's artwork
Zoom Range	$\pm 20\%$	Simulates drawings at different scales (small vs. large figures)
Shear Range	10%	Slight skewing simulates scanning angle variation
Brightness Range	[0.8 – 1.2]	Simulates lighting differences in photographs of paper drawings

Table 3: Data augmentation parameters applied to the training generator only.

5.2 Model Architecture and Hyperparameter Tuning

SVM baseline: HOG features were extracted from 64×64 grayscale images with 9 orientation bins, 8×8 pixels per cell, and 2×2 cells per block, producing a 1,764-dimensional feature vector per image. Train and validation features were merged (787 samples) for GridSearchCV fitting; the test set (142 samples) was held out. A 5-fold cross-validation grid search over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\text{scale}, \text{auto}, 0.001, 0.01\}$ (80 fits total, scored by macro F1) found the best configuration to be $C=100$, $\gamma=\text{scale} \rightarrow \text{CV F1} = 0.4680$. Higher C values consistently outperformed lower ones, meaning the SVM benefited from stricter enforcement of correct classification on the 1,764-dimensional HOG features.

CNN architecture: All three CNN models share the same custom classification head appended to their pretrained base: GlobalAveragePooling2D $\rightarrow$ BatchNormalization $\rightarrow$ Dense (256, ReLU) $\rightarrow$ Dropout (0.4) $\rightarrow$ Dense (128, ReLU) $\rightarrow$ Dropout (0.2) $\rightarrow$ Dense (4, Softmax). GlobalAveragePooling2D reduces the base's 3D feature maps to a 1D vector without flattening (which would be too large). BatchNormalization stabilizes activations. The two Dense layers compress and combine features before the final Softmax output, which produces a probability distribution over the four emotion classes. Dropout(0.4) and Dropout(0.2) provide regularization at each dense layer, preventing the model from over-relying on any single neuron — critical for

a small dataset of only 407 original images. The loss function is categorical cross-entropy and the optimizer is Adam.

CNN hyperparameters were fixed based on standard transfer learning best practices rather than exhaustive search (given compute constraints). Table 4 lists the key values and their justification.

Parameter	Value	Justification
Phase 1 LR / Epochs	1e-3 / 5	Standard starting LR for a new head; 5 epochs is enough for convergence before unfreezing the base
Phase 2 LR / Max Epochs	1e-5 / 20	100× smaller LR to fine-tune pretrained weights gently without catastrophic forgetting
Batch Size	32	Standard default; balances memory usage and gradient stability
Dropout	0.4 / 0.2	Heavier at Dense-256 (wider layer, more parameters to regularize), lighter at Dense-128
Fine-tuned Layers	Last 20 base layers	Earlier layers learn generic edges/textures (already useful); only later layers need domain adaptation

Table 4: Deep learning hyperparameters and justification.

5.3 Training Process and Callbacks

Each CNN was trained using a two-phase strategy. In Phase 1 (5 epochs, LR=1e-3), the base was fully frozen and only the head was trained. This prevents the pretrained weights from being corrupted before the head has learned a reasonable mapping. In Phase 2 (up to 20 epochs, LR=1e-5), the last 20 base layers were unfrozen for fine-tuning at a low learning rate to gently adapt the pretrained features to the drawing domain. The two phase histories were concatenated for learning curve analysis.

What happened per model: MobileNetV2 reached `val_accuracy=0.5500` in Phase 1 but declined immediately in Phase 2, with EarlyStopping triggering after 8 fine-tuning epochs (13 total). ResNet50 stagnated at `val_accuracy≈0.36` throughout both phases, with EarlyStopping also triggering at 8 fine-tuning epochs (13 total). VGG16 improved consistently — Phase 1 reached `val_accuracy=0.5643` and Phase 2 continued to a peak of 0.6714, running for all 19 epochs before EarlyStopping triggered.

Three callbacks were used throughout: EarlyStopping (`monitor=val_loss`, `patience=7`, `restore_best_weights=True`) to halt training and restore the best weights when validation loss stopped improving; ReduceLROnPlateau (`factor=0.3`, `patience=3`, `min_lr=1e-8`) to reduce the learning rate when training plateaued — observed to trigger for MobileNetV2 and ResNet50 in Phase 2; and ModelCheckpoint (`monitor=val_accuracy`, `save_best_only=True`) to save the best model weights at every improvement.

5.4 Evaluation Metrics

All models were evaluated on the same 142-image held-out test set — never seen during training or tuning. The following metrics were computed: Accuracy (overall correct predictions), Macro F1-Score (primary metric — unweighted average F1 across all four classes), Macro Precision, Macro Recall, and Test Loss (CNNs only). Per-class F1 scores, normalized confusion matrices (% per true class row), and one-vs-rest ROC curves with per-class AUC were also computed for deeper analysis.

Macro F1 was chosen as the primary metric because the test set classes are not perfectly balanced (Angry: 41, Fear: 32, Happy: 26, Sad: 43). A macro average weights all four classes equally regardless of size — appropriate when every emotion class matters equally, as it does in a psychological screening context where missing any emotion type has clinical consequences.

6. Results and Discussion

6.1 Overall Model Performance

Table 6 summarizes the test-set performance of all five implemented models. Results are reported in terms of accuracy, macro F1-score, macro precision, and macro recall, allowing a fair comparison across the four emotion classes.

Model	Test Accuracy	Macro F1	Macro Precision	Macro Recall
SVM (HOG)	48.59%	0.4871	0.4951	0.4815
MobileNetV2	50.00%	0.4906	0.5136	0.5387
ResNet50	32.39%	0.2609	0.3735	0.2876
VGG16 ★	60.56%	0.6068	0.6018	0.6376
Ensemble (Soft Voting)	60.56%	0.5992	0.5994	0.6319

Table 5: Final Model Comparison on Test Set (underline = Best Single Model)

VGG16 and the Soft-Voting Ensemble achieved the highest test accuracy (60.56%), while ResNet50 performed the poorest at 32.39%. The classical SVM baseline reached 48.59%, placing it between the weakest and strongest deep-learning models. MobileNetV2 marginally exceeded the SVM at 50.00%.

6.2 Generalization to Unseen Data

A key indicator of generalisation is the gap between training accuracy and test accuracy. VGG16 converged to a training accuracy above 80% by the end of fine-tuning, yet achieved 60.56% on the held-out test set – a gap of roughly 20 percentage points that signals moderate overfitting, which is expected given the small dataset size of 943 images across four classes. MobileNetV2 showed a similar pattern, peaking near 68% training accuracy with only 50% on the test set.

ResNet50 failed to generalise, with both training and validation accuracy remaining close to random chance (~35%) throughout all 25 epochs. Its residual skip connections, designed to help very deep networks, appeared to hinder learning from the relatively simple stroke-based features in children's drawings at this scale. The ensemble slightly smoothed individual model errors but

could not overcome ResNet50's drag on its aggregated probabilities, yielding the same top-line accuracy as VGG16 alone (60.56%) but with a lower macro F1 (0.5992 vs 0.6068), confirming that averaging a weak model into the pool can dilute ensemble gains.

6.3 Per-Class Performance

Tables 7 and 8 show the per-class breakdown for the SVM baseline and the best-performing Ensemble model respectively, providing insight into which emotional categories were easiest and hardest to distinguish.

Class	Precision	Recall	F1-Score	Support
Angry	0.44	0.44	0.44	41
Fear	0.31	0.34	0.33	32
Happy	0.64	0.54	0.58	26
Sad	0.59	0.60	0.60	43

Table 6: SVM (HOG) Per-Class Classification Report

Class	Precision	Recall	F1-Score	Support
Angry	0.68	0.61	0.64	41
Fear	0.52	0.41	0.46	32
Happy	0.59	1.00	0.74	26
Sad	0.61	0.51	0.56	43

Table 7: Ensemble (Soft Voting) Per-Class Classification Report

Across all models, Happy was consistently the easiest class to predict. The Ensemble achieved a perfect recall of 1.00 for Happy meaning every happy drawing in the test set was correctly identified at the cost of moderate precision (0.59), indicating some misclassifications of other classes as Happy. This may reflect that cheerful drawings tend to contain bright colours, smiling figures, and open compositions that form a visually distinct cluster.

Fear was the hardest class in both models, with the Ensemble reaching only F1 = 0.46. This is likely because fear-related drawings vary widely some depict dark, chaotic scenes while others

use muted colours with isolated figures making a single visual prototype difficult for the model to learn. The SVM's Fear F1 of 0.33 suggests that HOG edge-based features are especially ill-suited to capturing the diffuse visual signals of fear.

Angry and Sad were intermediate in difficulty. Both classes benefited most from the Ensemble's aggregation: Angry F1 rose from 0.44 (SVM) to 0.64 (Ensemble), and Sad F1 improved from 0.60 (SVM) to 0.56 (Ensemble), though the Sad improvement was less pronounced.

6.4 Confusion Matrices, ROC Curves, and Learning Curves

The confusion matrices below confirm the per-class narrative above. The Ensemble's matrix shows a dense diagonal for Happy and Angry, with most off-diagonal errors occurring between Sad and Fear the two classes with the most visual ambiguity. ResNet50's matrix is nearly random, with predictions spread almost uniformly across all four classes.

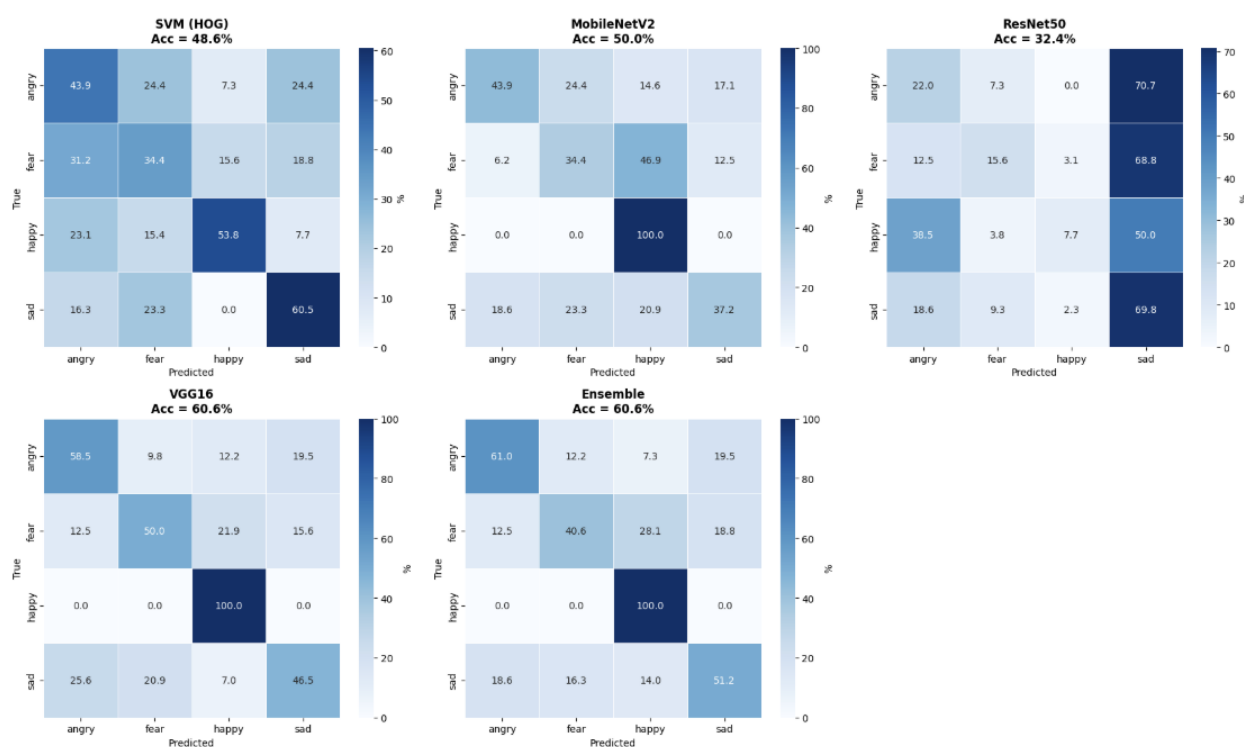


Figure 2: Confusion Matrices - All Models (% per row)

The ROC curves below were computed using a one-vs-rest strategy. VGG16 and the Ensemble achieved area-under-the-curve (AUC) values between 0.75 and 0.85 for Happy and Angry, while Fear yielded the lowest AUC (~ 0.65), consistent with its low F1. ResNet50 produced near-chance AUC values of approximately 0.50 for all classes, confirming its failure to learn any discriminative decision boundary.

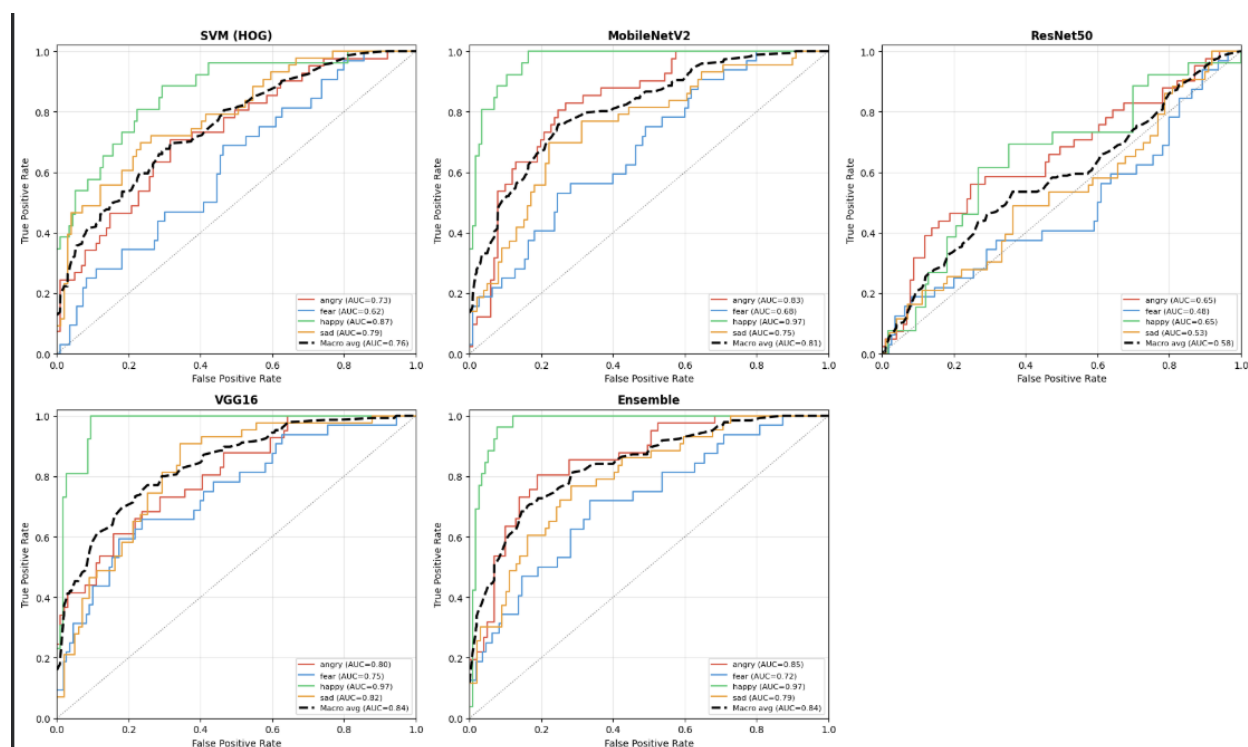


Figure 3: ROC Curves - All Models (One-vs-Rest)

The learning curves below reveal three distinct training dynamics. VGG16 showed steady convergence: both training and validation loss decreased over the 25 combined epochs, with validation loss plateauing after roughly epoch 15 a healthy sign of regularisation. MobileNetV2 exhibited early convergence in Phase 1 (frozen backbone, 5 epochs) but showed slight divergence during Phase 2 fine-tuning, where training loss continued to fall while validation loss stabilised, pointing to mild overfitting. ResNet50 displayed high, flat loss curves throughout, with validation loss slightly exceeding training loss from the very first fine-tuning epoch a pattern consistent with a model that was unable to adapt its ImageNet representations to the domain of children's drawings.

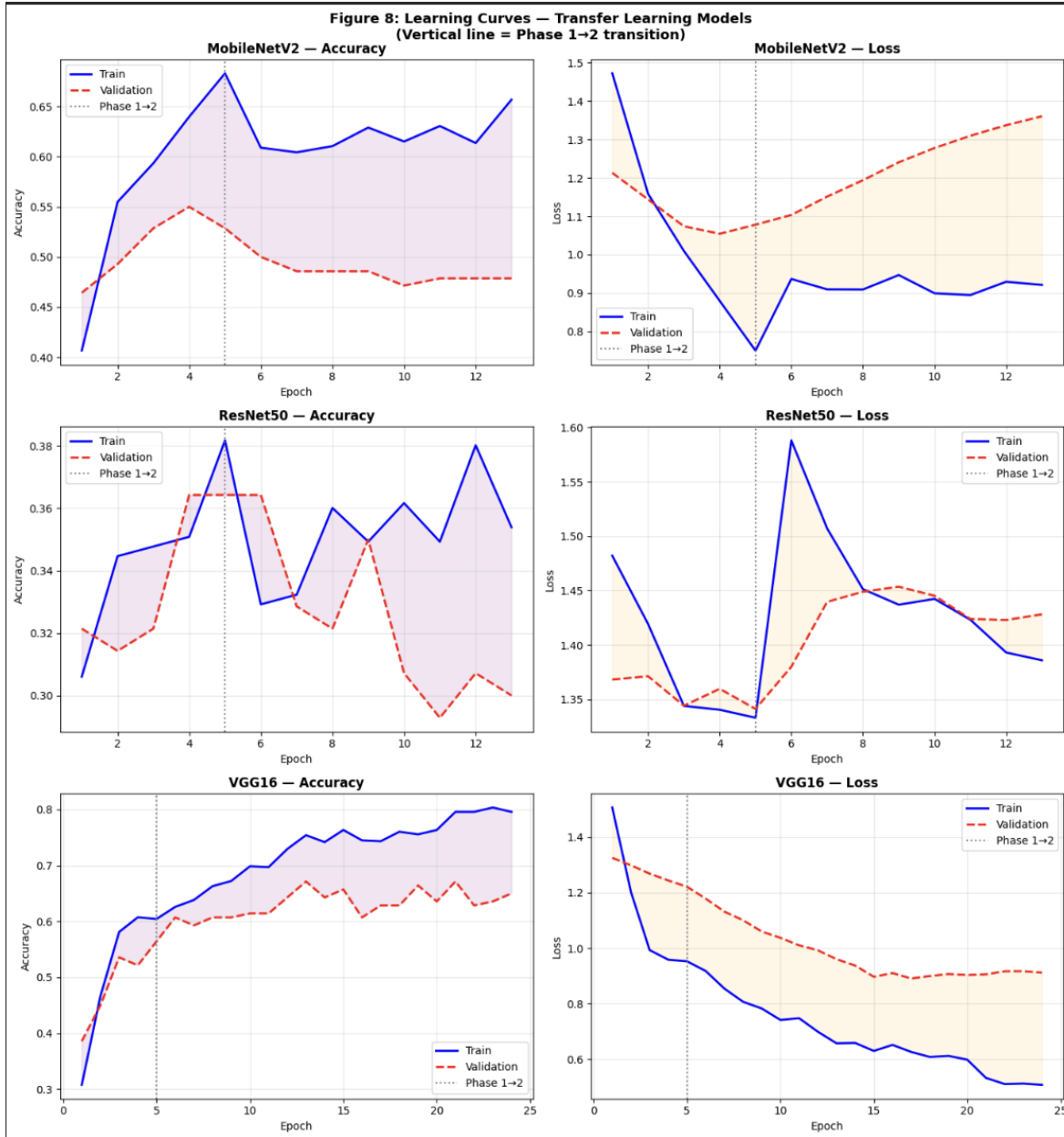


Figure 4: Learning Curves - Transfer Learning Models

6.5 Interesting Findings and Insights

Several noteworthy observations emerged from the analysis:

- VGG16 outperforms its larger, theoretically more powerful counterpart ResNet50. The sequential texture-learning capacity of VGG16's 13 convolutional layers appears better suited to children's drawing features primarily strokes, colour fills, and simple shapes

than the residual skip-connection architecture of ResNet50, which was designed for photographic images with complex spatial hierarchies.

- The classical SVM baseline (48.59% accuracy) was competitive with MobileNetV2 (50.00%) and vastly outperformed ResNet50 (32.39%). This suggests that when deep learning models fail to adapt, hand-crafted HOG features can remain a reliable fallback especially on small datasets.
- The Ensemble did not outperform VGG16 alone. Soft voting is most effective when component models are diverse and individually strong. Including ResNet50 in the ensemble introduced noise into the averaged probability vectors, demonstrating that ensemble composition quality matters more than ensemble size.
- Happy achieved 100% recall in the Ensemble, confirming the hypothesis that positive emotional drawings are visually distinctive characterised by warm colours, upward-curving lines, and rich figurative content making them easier for convolutional features to identify.
- The dataset augmented during training (647 effective training images after augmentation was applied per epoch) still represents a very small sample for four-class image classification. This is the primary bottleneck; the gap between training and test performance across all models reflects data scarcity rather than architectural limitations.

7. Conclusion

7.1 Key Findings

This project investigated whether supervised machine learning can reliably classify children's drawings into four emotional categories Happy, Angry, Sad, and Fear as a non-intrusive psychological screening tool. Five models were implemented and evaluated: a classical SVM with HOG features and four deep-learning approaches (MobileNetV2, ResNet50, VGG16, and a Soft-Voting Ensemble).

The central finding is that transfer-learned CNNs can extract psychologically relevant visual signals from children's drawings, but model architecture choice matters considerably. VGG16 achieved the best individual performance (60.56% accuracy, macro F1 = 0.61), substantially exceeding the classical baseline and demonstrating that deep visual representations offer genuine advantages over hand-crafted features for this domain. The Soft-Voting Ensemble matched VGG16's accuracy but failed to improve the F1-score, highlighting that ensemble gains depend on component model quality. The Happy class was the most reliably classified across all models, while Fear remained the most ambiguous – a finding that aligns with clinical observations that fear-related drawing features are less stylistically uniform than happy ones.

7.2 Challenges and Limitations

Several challenges and limitations were encountered during the project:

- **Small dataset size:** With only 407 original images (943 in the final expanded dataset), all models operated near the boundary of what deep learning can reliably learn. Data augmentation partially mitigated this, but the training-to-test accuracy gap across all models indicates persistent overfitting.
- **Label subjectivity:** The emotional labels in the dataset were assigned based on human annotation, which is inherently subjective. A drawing classified as 'Fear' by one annotator might be labelled 'Sad' by another, introducing label noise that degrades model performance.
- **ResNet50 domain mismatch:** ResNet50's residual architecture, optimised for photographic images with complex textures, failed to adapt to the simple stroke-based visual grammar of children's drawings, resulting in near-random performance and pulling down ensemble results.
- **Absence of metadata:** The model operates on raw pixel data only, without access to the child's age, drawing context, or any clinical annotations. Real psychological screening would benefit from multimodal inputs.

- Lack of clinical validation: Although classification accuracy provides a quantitative benchmark, agreement with expert psychologist labels – the gold standard in clinical practice – was not measured in this study.

7.3 Proposed Improvements and Future Directions

Based on the findings and limitations identified, several directions for future work are proposed:

- Dataset expansion: Collecting a larger, clinically validated dataset with hundreds of images per class would be the single highest-impact improvement. Collaboration with child psychologists to produce expert-annotated samples would also reduce label noise.
- Multimodal integration: Incorporating contextual metadata – such as the child's age, the drawing prompt, and any verbal explanations provided – could substantially improve classification by combining visual features with linguistic or demographic signals, as demonstrated by the KIDO dataset study referenced in the Related Works.
- Vision Transformers (ViT): Recent advances in Vision Transformers have shown strong performance on small-dataset image classification tasks through self-attention mechanisms that capture global context. Applying ViT or Swin Transformer architectures to this problem is a promising next step.
- Explainability (Grad-CAM): Generating class activation maps using Grad-CAM would allow visualisation of which regions of the drawing the model attends to when making a prediction, enabling clinical interpretability and building trust with practitioners.
- Better ensemble design: Replacing ResNet50 with an alternative model – such as EfficientNetB0 or a Vision Transformer – and applying a learned stacking meta-classifier rather than soft averaging could unlock genuine ensemble gains beyond what any individual model achieves alone.

8. References

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